Estimating Statistical Power using PRE and F-distributions.

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

keith.lohse@health.utah.edu

@keith_lohse

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Statistical decisions and types of errors.

Model C:

$y_i = \beta_0$

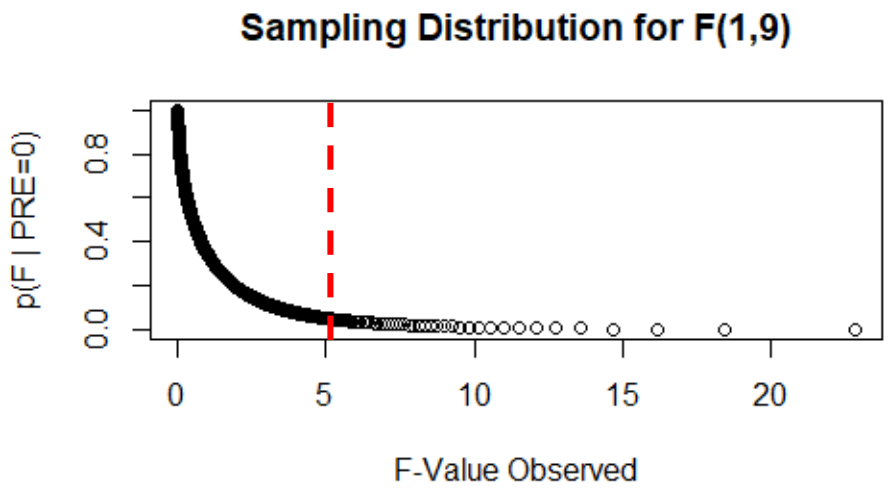
Model A:

$y_i = \beta_0 + \beta_1 x_i$

	True State of Nature	
Statistical Decision	*Model C Correct*	*Model C Incorrect*
"Reject Model C"	False Positive	True Positive
"Do not reject Model C"	True Negative	False Negative

- **False Positive (Type 1) Error**: We reject Model C when model C is ultimately correct.
- **False Negative (Type 2) Error**: We do not reject Model C when model C is ultimately not correct.

The false positive rate (α).



- **Given the null distributions,** we can find *critical* values of F.
- These values divide the F-distribution into different proportions, allowing us to make statements about the probability of F-observed.

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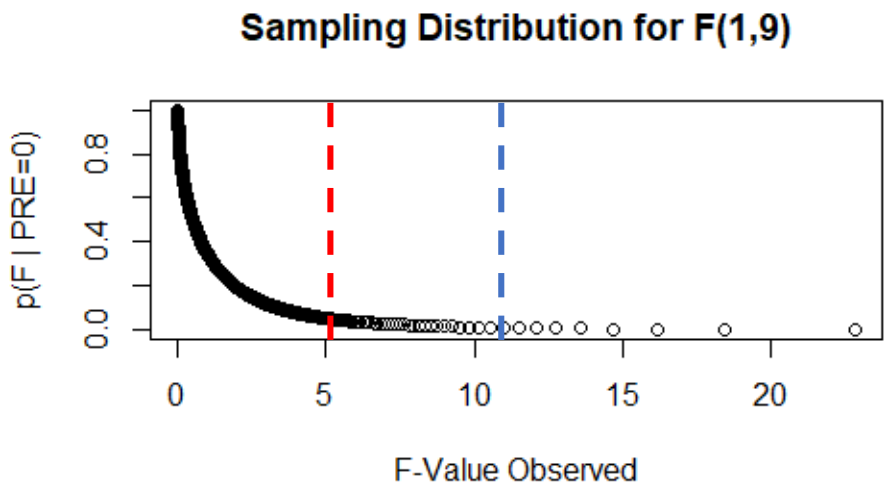

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$F_{(1,9);\alpha=0.01} = 10.56$

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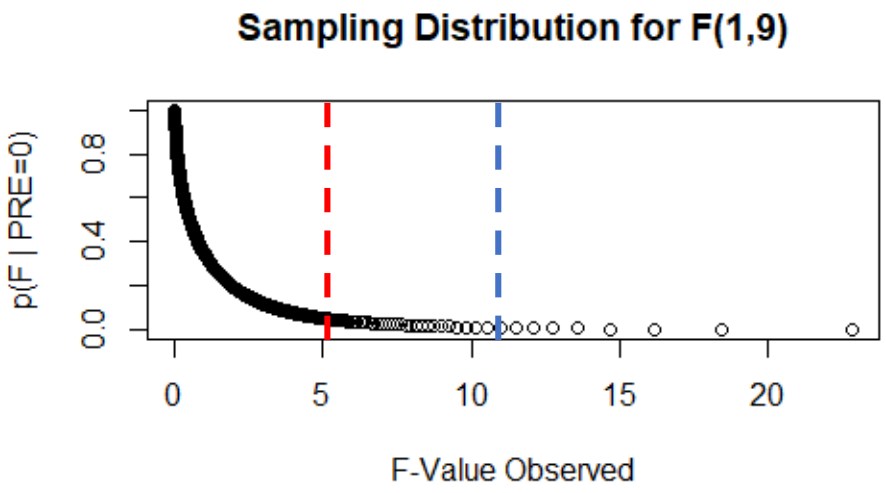


$F_{(1,9);\alpha=0.05} = 5.12$

$F_{(1,9);\alpha=0.01} = 10.56$

If $F_{obs} > F_{crit}$ then we would reject the null hypothesis.

The value of α determines how conservative we want to be about this decision. Lower α means fewer FP errors, but also more FN errors (all else being equal).

So, how can we avoid FP errors and reduce the change of FN errors?

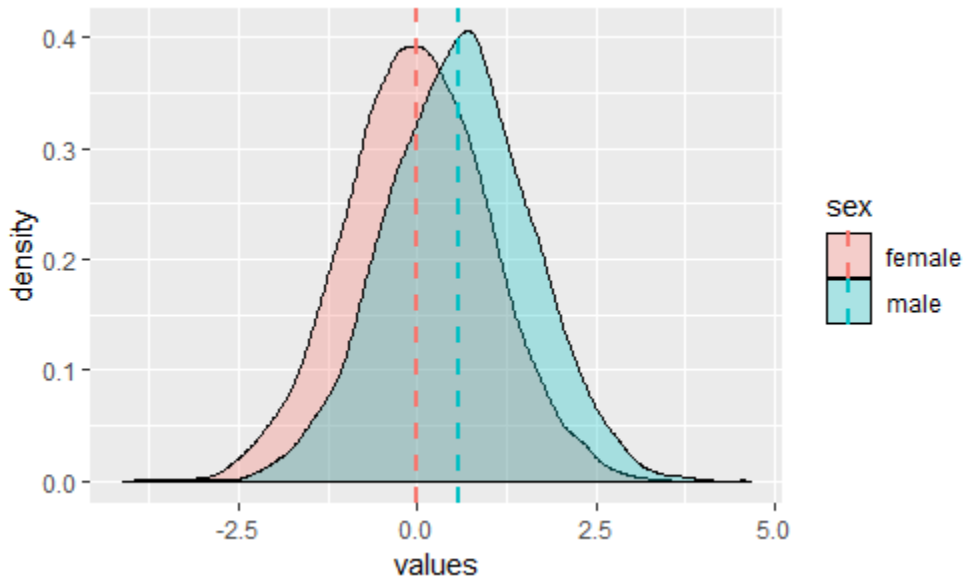
- **Given the null distributions,** we can find *critical* values of F.
- These values divide the F-distribution into different proportions, allowing us to make statements about the probability of F-observed.

Statistical Power

- Statistical power is the probability of rejecting the **null** hypothesis when a specific **alternative** hypothesis is assumed.
  - We know what the null F-distribution would look like, but we need make some informed assumptions about the effect we expect.
  - To distinguish this effect from the PRE we observe in our sample, the book refers to this value as the "***True PRE***" or $\boldsymbol{\eta}^2$.
  - This estimate should ideally be based on previous empirical data or a critical meaningful value (e.g., what is the smallest clinically meaningful change?).
  - When all else fails, there are "rule of thumb" effect-sizes that can be a useful starting point.

Statistical Power

- For instance, the difference in mass between adult men and women in the United States is about 0.56 standard deviations. (NHANES dataset. Heymsfield et al., 2014)



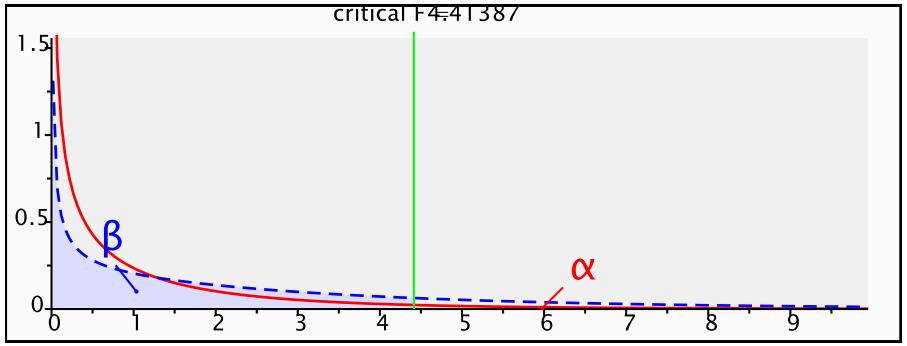
- By many, this would be considered a "moderate-sized" effect.

- This standardized difference translates into an $\boldsymbol{\eta^2 = 0.073}$.

Statistical Power

- Our power to detect this assumed difference thus depends on FP rate (α), the FN rate (β), and the sample size.

n/group = 10



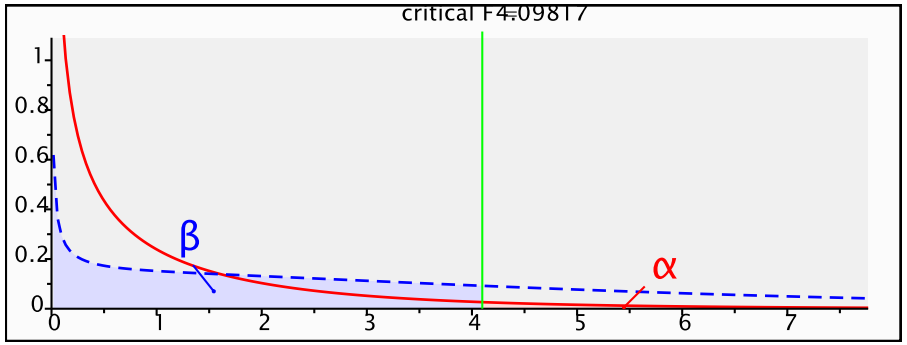
- If we have 10 people per group, we correctly conclude that men weigh more than women (on average) in only 22% of studies.

Given our assumed $\eta^2 = 0.073$ and $\alpha = 0.05$, our Power (1-β) = 0.22.

Statistical Power

- Our power to detect this assumed difference thus depends on FP rate (α), the FN rate (β), and the sample size.

n/group = 20



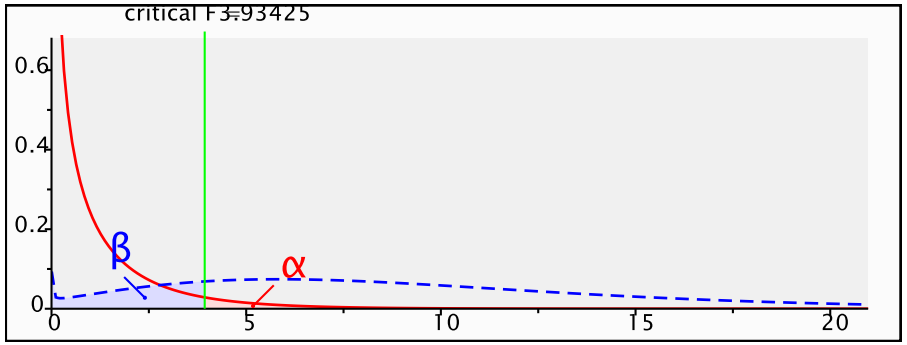
- If we have <u>20</u> people per group, we correctly conclude that men weigh more than women (on average) in only <u>41%</u> of studies.

Given our assumed $\eta^2 = 0.073$ and $\alpha = 0.05$, our Power (1-β) = 0.41.

Statistical Power

- Our power to detect this assumed difference thus depends on FP rate (α), the FN rate (β), and the sample size.

n/group = 52



- If we have <u>52</u> people per group, we correctly conclude that men weigh more than women (on average) in <u>81%</u> of studies.

Given our assumed $\eta^2 = 0.073$ and $\alpha = 0.05$, our Power (1-β) = 0.81.

Statistical Power

- Thus, if you think the effect you are trying to ascertain is more subtle than the fact that men weigh more than women, you probably want more than 50 people per group!

- Fortunately, there are other things we can do besides increasing *n* to improve statistical power.
  - Justify your α (e.g., if you are replicating an established effect or if the cost of a false positive is low, could α = 0.10?).
  - Reduce variability/error in your data (e.g. stratifying your sample or proper use of covariates can reduce unwanted variability).
  - Increase the effect-size (e.g., in a drug trial, do you need to use a subtle dose, or could you use a substantial dose?).